

Project Statement — SIH26103

MoSPI has a large amount of historical data related to major infrastructure projects, including project costs, expenditure, physical progress, milestones, timelines, delays and revisions.

Currently, this data is mainly used for monitoring, reporting and generating charts, but its historical patterns are not fully utilized to identify risks in ongoing projects.

Our proposed solution will use Machine Learning to learn patterns from completed and historical projects and apply those patterns to ongoing projects, enabling the system to predict the probability of cost overruns and time delays, identify the major factors contributing to the risk, assign an overall project-risk score, and generate early warnings when a project starts showing a pattern associated with future problems. This will help MoSPI move from reactive monitoring (“what has already happened?”) to proactive monitoring (“what is likely to happen next, and which project needs attention now?”).

In our language for learning and adding more suggestions

Project Statement — SIH26103

MoSPI ke paas bahut saare purane infrastructure projects ka data hai—jaise project cost, expenditure, physical progress, milestones, delays, completion dates aur cost revisions. Abhi is data ka use mainly reports, monitoring aur charts banane ke liye hota hai. Hum isi historical data ko Machine Learning ke training data ki tarah use karenge, jisse system purane projects mein ye identify karega ki kin patterns ya conditions ke baad cost overrun ya time delay hua tha. Phir jab koi current/ongoing project ka latest data system mein aayega, ML uske current pattern ko historical projects se compare karke cost overrun aur time overrun ki probability predict karega, risk score dega aur agar risk badh raha ho to early warning generate karega. Is tarah system sirf ye nahi batayega ki “project mein kya ho chuka hai”, balki ye bhi batayega ki “project mein aage kya problem aa sakti hai aur us problem ka signal abhi se dikh raha hai ya nahi.”

MoSPI ko actual main kaunsi problems face ho rahi hain?

1. Project ka actual progress time par aur accurately pata chalna
2. Delay hone ke baad pata chalna
3. Cost Overrun
4. Time Overrun
5. Different projects ko compare karna difficult hai
6. Problem ka actual reason identify karna
7. Huge amount of historical data ka proper use
8. Officer ko thousands of projects mein priority decide karni hoti hai

PS Face

Project ya site ki exact aur accurate information unke paas daily nahi pahuchti. Wo information unhe late milti hai, which was given by officers, tab tak work complete ho jata hai.

Problems

Officers contractors se commission lete hain aur wo jo report MOSPI ko dete hain, usme amount aur management sab badha kar bill dete hain, jisse project ki cost badh jati hai.

Persons / Roles jo pure project me hote hai

Contractor

Site Engineer jo bill lagata hai

AE — Assistant Manager

GM — General Manager

PM — Project Manager

MCE — Material Engineer

Sub Engineer jo measurement aur checking k liye site pe jata hai

QA/QC Engineer ye quality assurance OR quality check engineers hote hai

in ek pura easy flow

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CONSTRUCTION PROJECT HIERARCHY

TOP → BOTTOM

=====

+-----+	+-----+	+-----+
No	POSITION	MAIN RESPONSIBILITY
+-----+	+-----+	+-----+
01	Client / Owner	Project owner.
	Budget, approval & final	
	decisions.	
+-----+	+-----+	+-----+
02	PMC	Project Management
	Consultant. Client ki	
	taraf se monitoring.	
+-----+	+-----+	+-----+
03	Project Director	Overall project head.
	/ Head	Teams & contractors ko
	manage karta hai.	
+-----+	+-----+	+-----+
04	GM	General Manager.
	Project/region ki overall	
	responsibility.	
+-----+	+-----+	+-----+
05	PM	Project Manager.
	Overall execution, cost,	
	time & quality.	
+-----+	+-----+	+-----+
06	AM	Assistant Manager.
	PM ko assist & teams	
	coordinate karta hai.	
+-----+	+-----+	+-----+
07	Construction Manager	Actual construction
	execution ko manage karta	
	hai.	
+-----+	+-----+	+-----+
08	Project Engineer	Drawings, technical
	decisions & execution.	
+-----+	+-----+	+-----+
09	Site Engineer	Daily site construction
	work supervise karta hai.	
+-----+	+-----+	+-----+
10	Sub Engineer	Site measurements, work
	checking & coordination.	
+-----+	+-----+	+-----+
11	QA/QC Engineer	Material & construction
	quality check karta hai.	
+-----+	+-----+	+-----+
12	Planning Engineer	Project schedule, progress
	& delay tracking.	
+-----+	+-----+	+-----+
13	Billing Engineer	Quantities, contractor
	bills & RA bills.	
+-----+	+-----+	+-----+
14	Safety Engineer/HSE	Health, Safety &
	Environment. Accident	

| | | prevention & compliance. |

+---+-----+-----+-----+

| 15 | Surveyor | Levels, alignment, |

| | | coordinates & measurements. |

+---+-----+-----+-----+

| 16 | Supervisor | Labour & site work ko |

| | | supervise karta hai. |

+---+-----+-----+-----+

| 17 | Foreman | Workers ko daily work |

| | | assign & guide karta hai. |

+---+-----+-----+-----+

| 18 | Skilled Workers | Mason, carpenter, bar |

| | | bender etc. Actual skilled |

| | | construction work. |

+---+-----+-----+-----+

| 19 | Labour / Helpers | Physical construction & |

| | | support work. |

+---+-----+-----+-----+

=====

Planning Engineer — ye schedule banata hai.

Billing Engineer — jo bill banata hai.

→ Ab dekho jab sitepe kaam karna start karte hai to contractor jisne wo site/Project ka tender govt se liya hai ki wo bana ke dega

ab Pehle wo decide karta hai aur resource check karta hai ki uske paas kon konse resources hai is site ko complete karne k liye agar sab resources hai to ko khud is project ko complete karta hai

2nd option --->> wo petty contractor ko contractor de deta hai

ab petty contractor kya hai samjho

Petty Contractor = Main Contractor ke under chhote/specific construction work ko execute karne wala contractor.

Example:

Client/Owner → Main Contractor → Petty Contractor → Labour

Maan lo ek building ka project hai. Main Contractor ne poora project liya. Usne brickwork ka kaam ek chhote contractor ko de diya.

Woh petty contractor:

Masons/labour arrange karega

Brickwork karwayega

Daily work supervise karega

Apne assigned quantity/work ke hisaab se payment lega

Important difference

Petty Contractor ≠ Labour

Labour: sirf physical work karta hai.

Petty Contractor: labour ko arrange/manage karta hai aur ek particular work/package complete karne ki responsibility leta hai.

ab ye contractor decide karta hai contractor check karke uske resources and availability ko

ab petty contractor is type se contract lete hai

→ With material + tool + worker + maintenance . --- ye sab petty contractor ka

→ With material + worker + tool — --- ye sab petty contractor ka

→ With material + tool —.--- ye sab petty contractor ka

→ Tool + worker --- ye sab petty contractor ka

As per contractor kaam ke hisaab se decide kiya hota hai ki wo kitne worker, kitna material, kitne tools use karega. Uska material aur quantity + quality ke hisaab se kaam karega.

ab petty contractor kaam k hisab se payment leta hai contractor se aur ye corruption nhi kar parta hai kyuki iske under kuch nahi hota

kar sakta hai agar usne in type k contract liye ho

→ With material + tool + worker

→ With material + worker + tool

→ With material + tool

is me wo material bechh k wo contractor se aur paise mangega aur site banana me pura material use nahi karega

Ab yahan hota hai main contractor ka role.

Ab use bhi greed hota hai to jyada profit ke liye wo site engineers, material engineer, billing engineers, sub engineers, QA/QC ko commission deta hai.

To ye sab log quality check galat karte hai aur material ki quantity badha dete hai jo actual nahi hai, bill badha ke sab record govt ko dete hai

isse sabhi officers aur contractor ki double income hoti hai aur govt ka loss hota hai jo project ki cost ko badhata hai

Solution

1. Project ka actual progress time par aur accurately pata chalna

Government ke paas project ki information aati hai, lekin infrastructure project ground par continuously change hota rehta hai.

Example:

Project officially 65% complete report hua.

Lekin ground reality mein:

kuch sections complete hain

kuch sections mein work ruk gaya

bridge ka foundation complete hai but superstructure pending hai

material/site issue hai

land/clearance ka issue hai

Problem: Reported progress aur actual situation ke beech gap ho sakta hai.

To iske liye hum ek aisa AI automation app banaye.

Jisme project start se end tak pura daily ka data ya record ho, jisse supervisor jo kaam handle kare, ye check govt check kar sake.

aur is automation me ek page pura DAILY RECORD ka he hoga

aur usme filter ka option bhi hoga jaise ki unhe is date ka kaam dekhna hai to wo us date ka data dekhe ki us din kya hua tha

→ Is automation me hume ye dalna hai ki ye kaam road ka hai to isme puri us project ki details daali jaegi jaise size

location

cost

duration

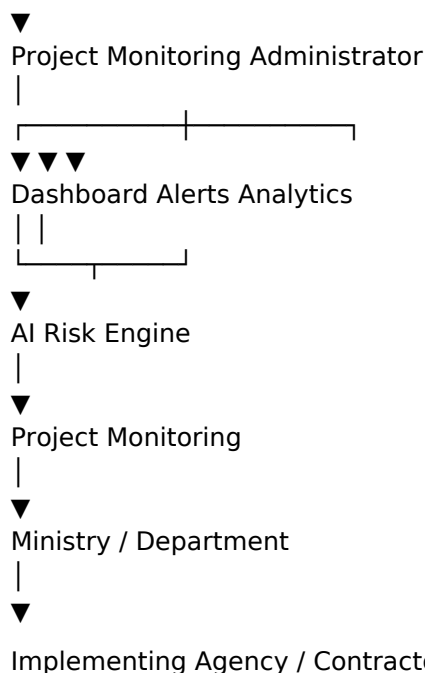
contractor

aur jo option wo add kare

ab ye project k andar automation chalega

MoSPI

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aur ye flow hoga aur jitney bhi officers hai wo connect honge par access sirf admin ko he honga to automation kya karega

→ Ab supervisor site location ki photo upload karega, to humara AI usko analyse karke batayega ki ye site ke pehle din ki photo hai.aur recommended question karega ki day one me kya kya hua kitna time laga ye sab aur ye sab optional hoga
Isme construction se pehle ki condition hogi.ye information apne daily report page me add hong

Officers ko dikhega jo bhi is projet me kaam karege par ise manage sirf MoSPI team he karegi jink paas iska access hai

To Automation ye karega

Day 1 — 1st Photo

Day 2 — Material unload

→ Uski photo + quantity batayega.+aur kaam kya hua ye bhi bataega aur ai khud puchega ki ye ye hua ya nahi kyuki AI ko pata hoga dusre din kya kaam hotaa hai
aur har ek material ka bill wahan upload hoga jise inspection team visit k time verify karke approve hogi kyuki bill amount badha kar diye ho to

Aur jo bhi ye image upload karega, usse AI puchega ki kitna material hai aur material ki bill receipt do. To ye bhi record me add hoga.

Day 3 — Site start, status dikhaega running

Ab jo image/photo dalega, use AI puchega kitna meter kaam hua, uski har ek angle se 4-5 photo + kitna material use hua aur used material ki photo

Jaise cement ke khali bag, kyunki corruption hota hai yahin par aur wo puchega ki quantity kitni thi, labour ki.

aur chahe to machinery ka fuel rent expenses optional hoga

kyuki ye verify bill engineer karega

aur aese he sabka flow bana ragega

Same process last day tak chalegi.

Aur is automation me saare officers connect honge to unko pass reminder jate rahega, jab jiska kaam hai jaise.

4 din hue to Material Engineer, Quality Engineer.

→ Aur daily site engineer jaega jo us daily record me apni geo tag photo aur uske kaam ki photos dalega to ye info bhi unhe pata chalegi aur daily record me rahega
isse ye pata chal jaega ki site engineer gaya hai

Sub Engineer ka jab bhi role hoga, wo us din jaake inspection karega aur photo geotag location

Same as case ke saare officers ke sath hoga aur

MoSPI team hai unke paas bhi iske through check karte rahege aur AI unhe reminder ya signal dega ki ye officer nahi gaya hai site visit

to unki team us officer se contact karegi aur ye signal officer ko bhi jaege

an is app me hum 2nd page bhi banaenge jo pura documentation ka hoga jisme sare officers sirf documents bhejege jo MoSPI ki team check karegi
aur in ka bhi access MoSPI team he karegi

→ Aur jo bhi officer hai, wo apna pura record bana ke usi app ke through sab documents bhej sakta.

→ Jisme sab docx bhejege pura record

Isse humare paas pura data hoga ki humare officers kab, kab gaye, jisse last me problem na ho.

2 Delay hone ke baad pata chalna

Ye sabse important problem hai.

Traditional monitoring mein:

Problem starts

↓

Progress slow

↓

Milestone miss

↓

Monthly/quarterly reporting

↓

Government notices delay

MoSPI ka goal ideally:

Problem starts

↓

System detects abnormal pattern

↓

⚠ EARLY WARNING

↓

Officer investigates

↓

Intervention

↓

Delay potentially reduced

Matlab: problem ko after-the-fact reporting ke bajay early warning mein convert karna.

solution

Ab dekho delay kyun-kya hota hai:

→ Workers kaam aana

→ Workers na aana

→ Tool na mil paana

→ Material / Resources ki unavailability

→ Liability

→ Ya site banate time kisi ki land me kuch hota hai to wo stay laga deta hai jaise construction k worker unke land me material rakhe ya use kare to wo person issue hai

Problem starts

↓

System detects abnormal pattern

↓

⚠️ EARLY WARNING

↓

Officer investigates

↓

Intervention

↓

Delay potentially reduced

ab ye sab info MoSPI ko late jaati hai

→ Aur machinery time pe na mil paana, isse site delay hota ha

to humara AI delay report karega jab tab site complete nahi hoti kyuki humare daily report wale part me sare reasons honge delay ke to iski bhi report and signals officers ko jaegi

ye daily site starting team se puchega ki kya update hai agar site on hai to photos aur status OFF hai to reasons to ye sab puri team dekhegi

aur in sabka handle MoSPI team se hoga aur isme project ka pura pura status statistics me dikh jaega

3. Cost Overrun

Infrastructure projects mein original sanctioned cost aur final/revised cost alag ho sakti hai.

Example:

Original Cost ₹500 Cr

↓

Scope/material/land/etc. issues

↓

Revised Cost ₹650 Cr

MoSPI ko sirf ye nahi pata hona chahiye ki:

"Cost ₹650 Cr ho gayi."

Better question:

"₹500 Cr se ₹650 Cr hone ka risk humein kab dikh gaya tha?"

Aur:

"Kaunse factors cost escalation ke early indicators the?"

iska bhi solution AI me hai jab material wagera sabka data AI daily report part me hoga aur sare bills usme honge aur worker machinery se leke sabka record usme hoga

aur wo ye cost ki report bhi khud bataega aur agar billing engineer jyada amount ka bill dalta hai to usse reasons puche jaenge then MoSPI team use approve degi

aur ye sab jab naya project MoSPI team banaegi uske andar hoga

isse corruption k chance bohot kam ho jaenge

Agar billing engineer kaam ka 20% progress ka bill dalta hai aur quantity galat dalta hai to daily record se AI match karke reasons then approve by MoSPI team

App batayegi ki itna kaam hua hai, isme itna material gaya hai aur AI ko bhi construction ka knowledge hoga, to wo bhi apna ek estimate calculate karke batayega.

Example: 25 meters ... to iski quantity itni honi hai, to ek expected material ... uski quantity ... billing engineer bill dega. Agar usme se over ... badhato hai, to usko bhi alert ...

aur ye sab cross check hoga team k through

aur ye option documentation me aaega aur agar koi wrong info submit karta hai to wo daily record se check karke reason puchke verify team approve hoke he hoga

4. Time Overrun

Same thing completion date ke saath.

Original completion:

March 2025

Extension:

September 2025

Further extension:

June 2026

Government ko ideally pehle hi signal milna chahiye:

"Current trajectory ke basis par March 2025 completion unrealistic hai."

Dekho maine abhi tak jo bhi detail batayi hai, usme se kaafi point match karenge

isme automaton kya karega

Jab new site banegi aur progress me changes honge, jo bhi time deadline ke according analyze karega, uske according estimate banega.

Aur deadlines ke according agar kaam nahi chal raha hai, to complete nahi hoga, to warning/reminder officers tak jayegi.

Aur iska main feature ye hai ki manager ko ye daily ke according ... dekhne ko milega ki isme kya reason aa raha hai.

5. Different projects ko compare karna difficult hai

Suppose:

50 highway projects

20 railway projects

15 bridge projects

30 power projects

Har project ka:

size

location

cost

duration

contractor

implementing agency

complexity

different hai.

Toh simple:

"Project A 70% complete hai"

kaafi meaningful nahi hai.

Better:

"Project A is 70% complete, but comparable projects at this stage average 82%."

Benchmarking yahan useful hai.

To hamara app me multi-site management rahega, jisme manager ek jagah alag-alag projects dekhega aur jis project me progress slow hogi, usko AI easily manage karega.

Hume is app me comparison page bhi dena hai, jahan se deadline, progress, cost, contractor sabhi details show karenge.

Jisme jo statistics me show karenge, wo statistics me show karenge ki kaunse project me slow hai aur kaunse project me ... hai. Is page me sabhi AI automation se judge hoga.

Example: pura automation se kaam kiya, jiska kitna kaam hua.

aur ye info bhi daily report se leke pata chalega + documents side se bhi kitna amount paid hai aur kitna project ko laga sab show hoga bas show hoga edit nahi hoga aur ye sabhi projects ka hoga jitney bhi MoSPI team banti hai projects usme sirf statistics se show karenge, edit ka option nahi rahega, like digital well being aur Screen Time jaise

6. Problem ka actual reason identify karna

Dashboard mein:

□ Project delayed

likhna easy hai.

Lekin officer ko chahiye:

Why?

Possible reasons:

- land acquisition
- forest/environment clearance
- utility shifting
- contractor performance
- tender issues
- funding/payment issues
- design changes
- scope changes
- material availability
- geological conditions
- weather
- law & order
- local issues
- inter-department coordination
- litigation

Yahi reason identification tumhare solution ka important part hona chahiye.

Agar koi problem humare daily routine aur reasons se match hoti hai to part 3 reason section me jaega part 3 isme reasons batane honge ki kya kyu kaise

Jaise material nahi hai, to material engineer ki responsibility material ko order karne ke liye signal ... ye reason dega, jisse upper officer discuss karke approve karega.

Kisi case me funding time pe nahi milti, to uska notice aur responsibility bhi billing engineer ke paas jayegi.

ye sab bhi MoSPI dekhegi

Aur site manager bhi daily check karega ki inme kya officer ke working/assignments lage.

phir isse problem ka actual reason usi platform par pata chal jaega team ko calling me time waste nahi karna hai

aur automation me ye bhi aayega feature hoga ki reason kaaya hai par officer ne response nahi kiya to iska bhi signal jaega aur in sabka record rahega ki time date aur kis liye bheja tha

7 Abhi historical data ka kya ho raha hai?
8 Hum usi data ka better use kaise karenge?
solution

1. Historical Data hota kya hai?

MoSPI ke paas purane infrastructure projects ka data hai.

Maan lo ek road project hai:

Project A

Year	Planned Progress	Actual Progress	Expenditure
------	------------------	-----------------	-------------

2022	20%	15%	₹100 Cr
------	-----	-----	---------

2023	40%	31%	₹250 Cr
------	-----	-----	---------

2024	65%	48%	₹430 Cr
------	-----	-----	---------

2025	85%	62%	₹620 Cr
------	-----	-----	---------

2026	100%	78%	₹850 Cr
------	------	-----	---------

Ab ye 5 saal ka data historical data hai.

Iske saath aur information bhi ho sakti hai:

Original project cost

Revised cost

Start date

Scheduled completion

Actual completion

Physical progress

Financial progress

Milestones

Delays

Extensions

Project status

Ministry

Sector

etc.

2. flow hai

Screenshot:

Historical Data

↓

Stored

↓

Reports

↓

Charts

Iska simple meaning:

Data collect hua → database mein store hua → usse reports bani → charts ban gaye.

For example, officer dashboard kholta hai:

Project A

Physical Progress = 78%

Expenditure = ₹850 Cr

Status = Ongoing

Ya report mein:

2022 → 15%

2023 → 31%

2024 → 48%

2025 → 62%

2026 → 78%

Graph bana diya.

Isme problem kya hai?

Data kharab nahi hai.

Problem ye hai ki system primarily "WHAT HAS HAPPENED?" batata hai.

Lekin administrator ko ek aur important question chahiye:

"WHAT IS LIKELY TO HAPPEN NEXT?"

3. Ab same data ko ML mein daalo

Screenshot ka second flow:

Historical Data

↓

ML

↓

Patterns

↓

Risk Prediction

↓

Early Warning

Iska matlab bahut simple example se samjho.

Hamare paas 10,000 completed/past projects ka data hai.

ML un projects ko dekhta hai aur patterns learn karta hai.

Suppose ML notice karta hai:

Jin projects mein expenditure bahut fast increase hua + physical progress expected se continuously kam rahi + milestones baar-baar delay hue, unmein future mein cost/time overrun ka chance zyada tha.

Ye pattern hai.

4. Ab current project ko dekho

Maan lo current Project X:

Expected Progress = 70%

Actual Progress = 52%

Expenditure = 68%

Milestones:

3 delayed

Human officer bhi bol sakta hai:

"Lagta hai project slow hai."

But ML historical projects se compare karke bol sakta hai:

"Projects with this type of progress-expenditure pattern historically had a high probability of schedule overrun."

Ab system:

Current Project

↓

Historical patterns se comparison

↓

Risk calculation

↓

⚠ Early Warning

5. "Risk Prediction" ka actual meaning

Ye future ko 100% predict karna nahi hai.

Hum ye nahi bolenge:

"Project definitely 8 months late hoga."

Instead:

"Current data ke basis par project ke 8+ months delay hone ki estimated probability 82% hai."

Example:

Project X

Cost Overrun Risk 74%

Time Overrun Risk 86%

Overall Risk 89/100

☐ CRITICAL

6. Sabse important: Early Warning

Yahi is poore concept ka main benefit hai.

Suppose project abhi officially "Ongoing" hai.

Abhi delay officially declare nahi hua.

Lekin ML bolta hai:

Risk Trend

April 38

May 44

June 51

July 63

August 78

September 87

System officer ko alert kare:

⚠ Early Warning: Project risk has increased significantly.

Primary indicators:

Physical progress below planned trajectory

Expenditure progressing faster than physical work

Multiple milestone delays

Ab officer problem officially huge hone se pehle investigate kar sakta hai.

7. Ek real-life road example

Maan lo ek highway project hai.

Initially:

Project Cost = ₹1,000 Cr

Duration = 3 years

First year:

Expected physical progress = 30%

Actual physical progress = 24%

Expenditure = 32%

Abhi koi massive problem nahi dikhti.

Second year:

Expected progress = 65%

Actual progress = 47%

Expenditure = 68%

Yahan ek interesting signal hai:

₹68% paisa spend ho chuka hai, but physical work only 47% hai.

Historical data mein ML ko pata chal sakta hai ki aise patterns wale projects mein future cost/time overrun ka risk historically higher raha hai.

Therefore:

Current Data

↓

Historical Projects se comparison

↓

Pattern detected

↓

High Risk

↓

⚠️ EARLY WARNING

8. Lekin ek bahut important baat

Yahan hum sirf historical data se prediction nahi karenge.

Historical data humein learning material dega.

Think of it like this:

Human example

Agar tumne 1,000 road projects dekhe hain, tum experience se bol sakte ho:

"Is type ka pattern generally problem create karta hai."

ML bhi kuch similar karta hai.

1000s of old projects

↓

ML learns patterns

↓

New/current project

↓

Pattern matching

↓

Risk probability

Difference ye hai ki ML bahut saare variables simultaneously analyze kar sakta hai.